Probability Theory: Review

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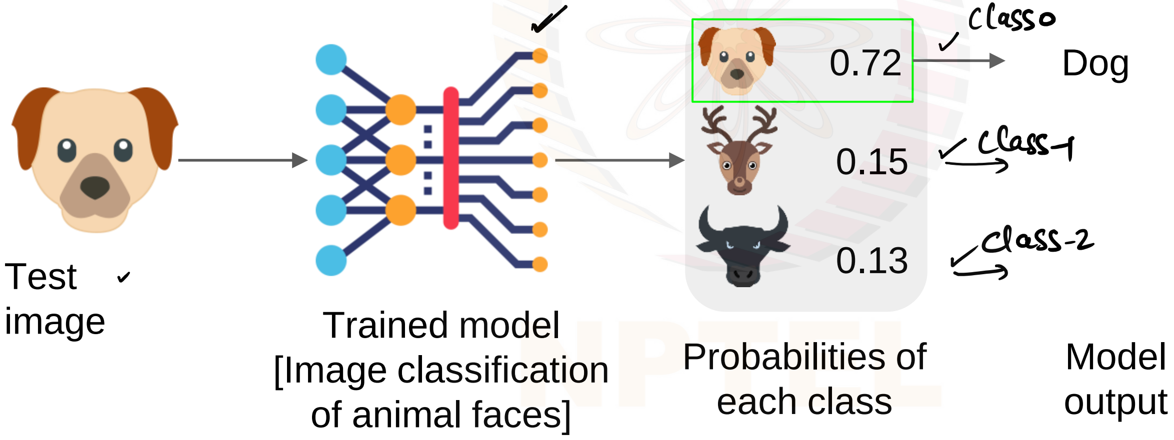
Probability Theory: Review

(Basics of Probability)

Motivation

- Probability theory is the foundation of many ML algorithms (E.g., Naïve Bayes)
- Provides a rigorous framework for modeling uncertainty & making predictions based on probabilities
- Example:
  - Classification probability



Learning Objectives

- Why do we study?
- Definition
- Elements of Probability
    - Examples
- Types of Events
- Axioms & Laws
- Total Probability Theorem
- Bayes Theorem
- Bernoulli Trials

Reference Text.: Stark, Henry, and John William Woods. "Probability and random processes with applications to signal processing." (2002).

Why do we study?

- Describes a model of a physical system by providing a quantification of uncertainty by repetitive experiments
- Examples:
  - Flip a coin. Did it land with heads or tails facing up?
  - How long do you wait for arrival of a bus?
  - In ML lecture, how many students will sit in the last row?

Definition

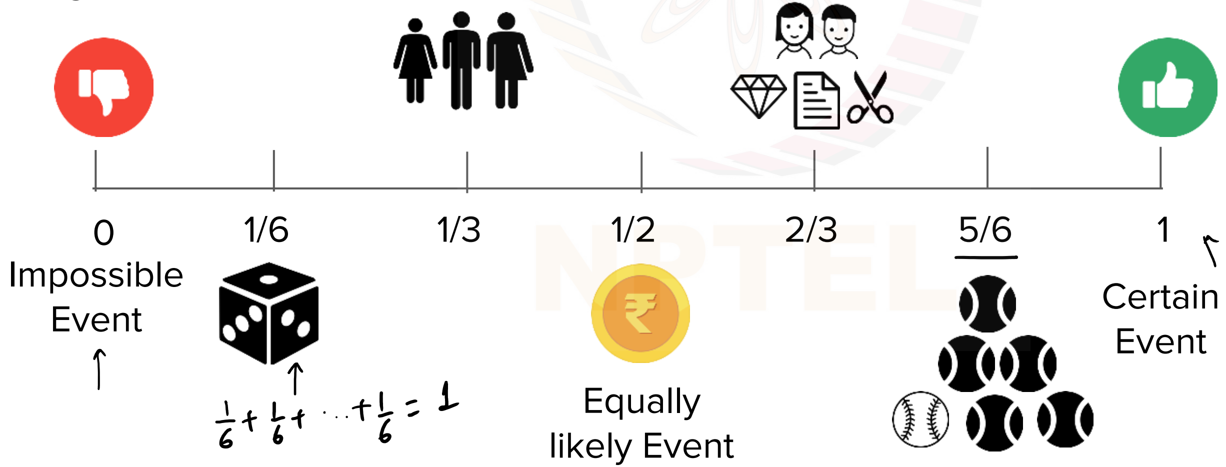
- Describes a model of a physical system by providing a quantification of uncertainty by repetitive experiments
- Probability refers to the measure of the likelihood or chance of an event occurring

$$Probability(Event\ A) = \frac{Number\ of\ favorable\ outcomes\ of\ A}{Total\ number\ of\ possible\ outcomes}$$

- Range: [0,1]

$P(H) = \frac{40}{100}$

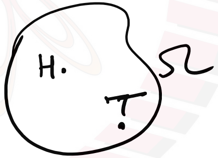
$\frac{1}{2} + \frac{1}{2}$

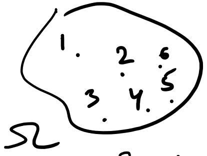


$\frac{1}{6} + \frac{1}{6} + \cdots + \frac{1}{6} = 1$

Elements of Probability

- **Random Experiment**: An experiment that has many outcomes and, in any trial of this experiment, the outcomes are uncertain
  - Tossing a coin
  - Throwing a dice
- **Outcome/ Elementary Event**
- **Sample Space (Ω) or the Universal Set**: The set of all possible outcomes of the experiment
  - Tossing a coin: $\Omega = \{H, T\}$
  - Throwing a dice: $\Omega = \{1,2,3,4,5,6\}$





- **Event (E)**:
  - Subsets of the sample space

$\phi = \{\}, \Omega, \{H\}, \{T\}$

$\{2,4,6\}$
$\{1,3,6\}$
$\{1,2\}$
$\{4,5,6\}$

- **Odds of an Event**:
  - Odds in favor of an event $= \dfrac{\text{Number of favorable outcomes}}{\text{Number of unfavorable outcomes}}$
  - Odds against an event $= \dfrac{\text{Number of unfavorable outcomes}}{\text{Number of favorable outcomes}}$

Example: Elements of Probability

- **Experiment**: Toss a coin twice

$\Omega = \{H, T\}$ $2^2 = 4$

$\phi, \Omega, \{H\}, \{T\}$

- **Sample space**:
  - $\Omega = \{HH, HT, TH, TT\}$
- **No. of possible events**: $\underline{2^n}$, where n is the number of possible elementary events or outcomes
  - All possible subsets of Ω constitutes the **Power Set**: Ω, {HH}, {HT}, {TH}, {TT}, {HH, HT}, {HH, TH}, {HH, TT}, {TH, HT}, {HT, TT}, {TH, TT}, {HH, HT, TH}, {HH, TH, TT}, {HH, HT,TT}, {HT, TH, TT}, ϕ

- **Concept of Sigma Field**: Consider a universal set Ω and the collection of its subsets. Let A, B,… denote the subsets belonging to Ω. This collection of subsets forms a field $\mathcal{F}$ if
  1) $\phi \in \mathcal{F}$ and $\Omega \in \mathcal{F}$
  2) If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$
  3) If $A \in \mathcal{F}$ and $B \in \mathcal{F}$, then $A \cup B \in \mathcal{F}$ and $AB \in \mathcal{F}$

  A sigma field is a field that is closed under any countable set of unions, intersections and combinations of events.

Axioms of Probability

Axioms: Basic rules that govern the behavior of probabilities

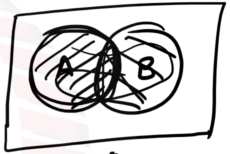
1) $P(\Omega)=1$ ✓ certain event

2) $P(\phi)=0$

3) $0 \leq P(A) \leq 1$

4) $P(AUB)= P(A) + P(B) - P(AB)$ if A & B are not disjoint

$P(A \cap B)$



$A \cap B = \phi$

$P(A \cap B) = 0$

$= P(A) + P(B)$ if A & B are disjoint

Laws of Probability

Laws: Derived rules that follow from the axioms and help compute probabilities

- **Addition law**: the probability of the union of two events is the sum of their probabilities minus the probability of their intersection.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- **Multiplication law**: the probability of the intersection of two independent events is the product of their probabilities (if A and B are independent)

$$P(A \cap B) = P(A) * P(B) = P(A)P(B)$$

- **Complement law**: the probability of the complement of an event is 1 minus the probability of the event.

$$\bar{A} = \Omega - A$$

$$P(\bar{A}) = 1 - P(A)$$

$$P(\bar{A}) = 1 - P(A)$$

- **Conditional Probability**: The probability of an event occurring given that another event has already occurred.

$$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{P(AB)}{P(B)}$$

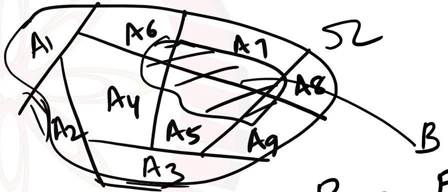
if A & B are independent

$$P(AB) = P(A)P(B)$$

$$P(A/B) = \frac{P(AB)}{P(B)} = \frac{P(A)\cancel{P(B)}}{\cancel{P(B)}} = P(A)$$

Total Probability Theorem [Law of Total Probability]

Given a set of mutually exclusive and exhaustive events $\{A_1, A_2, ..., A_n\}$ (i.e., exactly one of these events must occur), and an event B over this space, then the probability of B can be calculated as the sum of the probabilities of B given each of the A_i events, weighted by the probability of each A_i occurring.



$n = 9$

$\rightarrow A_i \cap A_j = \phi \quad \forall i \neq j$

$\Omega = \bigcup_{i=1}^{n} A_i$

$B = B\Omega$

$= B\left(\bigcup_{i=1}^{9} A_i\right)$

$= \bigcup_{i=1}^{n} BA_i$

$P(B) = P(BA_1) + P(BA_2) + \ldots\ldots\ldots + P(BA_n)$

$= P(B/A_1)P(A_1) + P(B/A_2)P(A_2) + \ldots\ldots\ldots + P(B/A_n)P(A_n)$

$A_i \cap A_j = \phi$

$BA_i \cap BA_j = \phi$

$\Rightarrow$

$P(B/A_1) = \dfrac{P(BA_1)}{P(A_1)}$

$P(BA_1) = P(B/A_1)P(A_1)$

Bayes' Theorem

"The probability of an event A given that event B has occurred is equal to the probability of event B given that event A has occurred, multiplied by the probability of event A occurring, and divided by the probability of event B occurring."

posterior

likelihood

apriori prob.

(prior)

$$P(A|B) = \frac{P(B|A) * P(A)}{P(B)}$$

evidence

$$P(AB) = P(A/B)\,P(B)$$
$$= P(B/A)\,P(A)$$

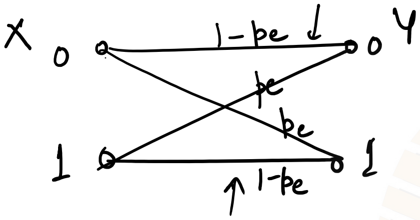
Example: Binary Symmetric Channel (BSC)

$P(X=0)=\frac{1}{2}$

$P(X=1)=\frac{1}{2}$

apriori prob.

Tx



Channel

Rx

$$p_e \equiv P(Y=1/X=0) = P(Y=0/X=1)$$ transition prob.

$$P(X/Y) = \frac{P(Y/X)\,P(X)}{P(Y/X=0)\,P(X=0) + P(Y/X=1)\,P(X=1)}$$

aposteriori prob.

$$P(X=0/Y=0) = \frac{P(Y=0/X=0)\,P(X=0)}{P(Y=0/X=0)\,P(X=0) + P(Y=0/X=1)\,P(X=1)}$$

$$= \frac{(1-p_e)\frac{1}{2}}{(1-p_e)\frac{1}{2} + p_e \times \frac{1}{2}}$$

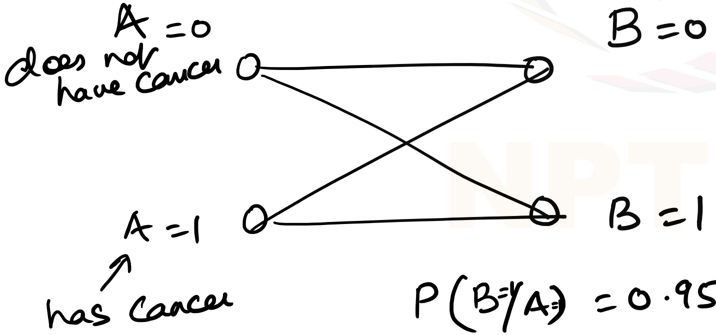
$$1 - P(X=0/Y=0) = P(X=1/Y=0)$$

Try it!

Q. Suppose that a new cancer test has been devised by a company. Let us define events as

A = event that the person has cancer
B = event that the test states that the person has cancer

It is known that the prevalence of cancer in the population is 0.5%. The test correctly recognizes 95% subjects of the positive class (having cancer) and correctly recognizes 95% subjects of the negative class (not having cancer), i.e., $P(B/A) = 0.95 = P(B^c/A^c)$. Is it a good test?



$B=0$ ← test states that the person does not have cancer

test states that the person has cancer

$P(B=1/A=1) = 0.95$

$P(B^c/A^c) = 0.95$

$P(A=1) = 0.005$

Example

$$P(A=1/B=1)$$

$$P(A=0/B=0)$$

Bernoulli Trial & Binomial Experiment

Bernoulli Trial: Experiment with a binary outcome: Success (S) or Failure (F)

$\Omega=\{S,F\}$ No. of possible outcomes = 2

$q = 1-p$

$P(S) = p$; $P(F) = q = 1-p$

$\mathscr{F} = \{\Omega, \{S\}, \{F\}, \phi\}$

Binomial Experiment: Perform the Bernoulli trial experiment n times $(n > 1)$

$\underbrace{\Omega \times \Omega \times \ldots\ldots\ldots \times \Omega}_{n}$ No. of possible outcomes = 2^n

$2^3 = 8$

Example: Say a coin is tossed 3 times (n=3).

Label Success = receiving head; Failure= receiving tail.

Compute the probability of receiving 2 heads and one tail.

Binomial Experiment Cont…

Compute the probability of receiving 2 heads and one tail.

Event E= Receiving 2 heads and one tail = (2 successes and one failure)

= {HHT, HTH, THH}

pqp

$P(HHT) = p^2q$

$P(HTH) = p^2q$

$P(THH) = p^2q$

$P(E) = 3p^2q \quad = {}^3C_2\, p^2q$

In general, the probability of any event A with k successes and (n-k) failures out of (n) Bernoulli trials in a binomial experiment is given by:

$$P(A) = {}^nC_k\, p^k q^{n-k} = b(k; n,p)$$

and

$$\sum_{k=0}^{k=n} b(k; n, p) = 1$$

Try it!

(a) Compute the probability of having k or fewer successes

$$\sum_{i=0}^{k} b(i; n, p) = {}^nC_0\, p^0 q^n + {}^nC_1\, p^1 q^{n-1} + \dots + {}^nC_k\, p^k q^{n-k}$$

(b) Compute the probability of having at least k successes

$$\sum_{i=k}^{n} b(i; n, p) = {}^nC_k\, p^k q^{n-k} + {}^nC_{k+1}\, p^{k+1} q^{n-k-1} + \dots + {}^nC_n\, p^n q^0$$

(c) Compute the probability of more than k but less than or equal to j successes (given j > k)

$$\sum_{i=k+1}^{j} b(i; n, p)$$

Try it!

Q. Five missiles are fired against an enemy ship in the ocean. At least two missiles should hit the ship to sink it. All missiles are on the correct trajectory, but the ship's defense system can destroy the missile with probability 0.95.

$p(\text{success})$ = missile hit the ship = 0.05

$p(\text{failure})$ = " is not able to hit = 0.95

Binomial : $n = 5$, $p = 0.05$

(a) What is the probability that the enemy ship is sunk for sure?

$$\sum_{i=2}^{5} {}^{n}C_i \, p^i q^{n-i} = \sum_{i=2}^{5} {}^{5}C_i \, p^i q^{n-i}$$

$$= {}^{5}C_2 (0.05)^2 (0.95)^3 + \dots + {}^{n}C_5 (0.05)^5$$

(b) What is the probability that the ships stays afloat?

$$1 - P(\text{ship sinks}) = {}^{5}C_0 \, p^0 q^n + {}^{5}C_1 \, p^1 q^{n-1}$$

Summary

Random experiment- the concept of a sample space

Concept of probability and axioms of probability

Types of events and their probabilities:

- **Certain Event:** that is guaranteed to occur Ω
- **Impossible Event**: event that cannot occur ϕ
- **Equally Likely Events**: that have an equal probability of occurring
- **Mutually Exclusive Events**: that cannot occur at the same time
- **Independent Events**: where the occurrence of one event does not affect the probability of the occurrence of the other event
- **Exhaustive Events**: that cover the entire sample space

Total probability theorem and Bayes theorem

- Conditional probability
- Transition probabilities, apriori and aposteriori probabilities

Bernoulli trial and Binomial experiment

Examples